

C. F. Gauss

Theoria residuorum biquadraticorum. Commentatio secunda English translation of Articles 66–76

Art. 66

For the number -3 we omit the distinction between moduli of the first and of the second kind, since the result shows that here it is superfluous. Thus there corresponds

character	moduli
0	$-1 + 6i, -1 - 6i, -7, -5 + 6i, -5 - 6i, -11, 11 + 6i, 11 - 6i$
1	$-1 - 2i, 1 - 4i, -5 + 2i, 5 + 4i, 7 + 2i, 5 - 8i, -1 + 10i, -7 - 8i, -11 - 4i, 7 - 10i$
2	$3 + 2i, 3 - 2i, -3 + 8i, -3 - 8i, 9 + 4i, 3 + 10i, 3 - 10i$
3	$-1 + 2i, 1 + 4i, -5 - 2i, 5 - 4i, 7 - 2i, 5 + 8i, -1 - 10i, -7 + 8i, -11 + 4i, 7 + 10i$

Reducing these moduli to least residues modulo 3, we see that those to which character 0 corresponds are partly $\equiv 1$ and partly $\equiv -1$; those whose character is 1 become either $\equiv 1 - i$ or $\equiv -1 + i$; those whose character is 2 become either $\equiv i$ or $\equiv -i$; finally, those to which character 3 belongs are either $\equiv 1 + i$ or $\equiv -1 - i$. From this induction we therefore infer that the character of the number -3 with respect to a modulus which is a primary prime among its associates is identical with the character of this same number when 3, or, what comes to the same thing, -3 , is considered as modulus.

Art. 67

By a similar induction carried out about other prime numbers, we find that the numbers $3 \pm 2i, -1 \pm 6i, 7 \pm 2i, -5 \pm 6i$, etc., furnish theorems similar to the one at which we arrived in Article 65 with respect to the number $-1 + 2i$; on the other hand, the numbers $1 \pm 4i, 5 \pm 4i, -3 \pm 8i, 9 \pm 4i$, etc., behave in the same way as the number -3 .

Induction therefore leads to an exceedingly elegant theorem, which, after the model of the theory of quadratic residues in the arithmetic of real numbers, may be called the *fundamental theorem of the theory of biquadratic residues*, namely:

Let $a + bi$ and $a' + b'i$ denote distinct prime numbers, each primary among its associates, that is, congruent to unity modulo $2 + 2i$. The biquadratic character of the number $a + bi$ with respect to the modulus $a' + b'i$ will be identical with the character of the number $a' + b'i$ with respect to the modulus $a + bi$ if either both of the numbers $a + bi, a' + b'i$, or at least one of them, belongs to the first kind, that is, is congruent to unity modulo 4. On the other hand, those characters will differ from one another by two units if neither of the numbers $a + bi, a' + b'i$ belongs to the first kind, that is, if each is congruent to $3 + 2i$ modulo 4.

Notwithstanding the great simplicity of this theorem, however, its proof must be counted among the most deeply hidden mysteries of higher arithmetic, so that, at least as matters now stand, it can be unravelled only by very subtle investigations, which would far exceed the limits of the present memoir. We therefore reserve for the third memoir the publication of this proof,

as well as the development of the connection between this theorem and the results which at the beginning of this memoir we had begun to establish by induction. As a kind of crown to the present discussion, however, we shall give here what is needed for the proof of the theorems proposed in Articles 63 and 64.

Art. 68

We begin with prime numbers $a + bi$ such that $b = 0$ (the third species of Article 34). Thus, in order that the number be primary among its associates, a must be a negative real prime number of the form $-(4n + 3)$; for it we shall write $-q$, such as -3 , -7 , -11 , -19 , and so on.

Let λ denote the character of the number $1 + i$ when that number is taken as modulus. We must have

$$i^\lambda \equiv (1 + i)^{\frac{1}{4}(qq-1)} \equiv 2^{\frac{1}{8}(qq-1)} \cdot i^{\frac{1}{8}(qq-1)} \pmod{q}.$$

But it is known that 2 is a quadratic residue, or a quadratic non-residue, of q according as q is of the form $8n + 7$, or of the form $8n + 3$. Hence we infer that, in general,

$$2^{\frac{1}{2}(q-1)} \equiv (-1)^{\frac{1}{4}(q+1)} \equiv i^{\frac{1}{2}(q+1)} \pmod{q},$$

and therefore, raising to the power whose exponent is $\frac{1}{4}(q + 1)$,

$$2^{\frac{1}{8}(qq-1)} \equiv i^{\frac{1}{8}(q+1)^2} \pmod{q}.$$

The preceding equation therefore takes the form

$$i^\lambda \equiv i^{\frac{1}{8}(q+1)^2 + \frac{1}{8}(qq-1)} \equiv i^{\frac{1}{4}(qq+q)} \pmod{q},$$

from which it follows that

$$\lambda \equiv \frac{1}{4}(qq + q) \equiv \frac{1}{4}(q + 1)^2 - \frac{1}{4}(q + 1) \pmod{4},$$

or, since $\frac{1}{4}(q + 1)^2 \equiv 0 \pmod{4}$,

$$\lambda \equiv -\frac{1}{4}(q + 1) \equiv \frac{1}{4}(a - 1) \pmod{4}.$$

This is precisely the theorem of Article 63 for the case $b = 0$.

Art. 69

Far more difficult are those moduli $a + bi$ for which b is not 0 (numbers of the fourth species in Article 34), and several investigations must be put in place first. We shall denote by p the norm $aa + bb$, which will be a real prime number of the form $4n + 1$.

Let S denote the complex of all simply least residues modulo $a + bi = m$, zero excluded; thus the number of entries contained in S is $p - 1$. Let $x + yi$ denote, indefinitely, a number of this system, and put

$$ax + by = \xi, \quad ay - bx = \eta.$$

Then ξ and η will be integers contained exclusively between the limits 0 and p . In the present case, where a and b are relatively prime, the formulas of Article 45, namely

$$\eta \equiv k\xi, \quad \xi \equiv -k\eta \pmod{p},$$

show that neither of the numbers ξ , η can be 0, unless the other vanishes at the same time, and hence $x = 0$, $y = 0$, a combination that we have already excluded. Thus the criterion for a number $x + yi$ contained in S consists in the positivity of the four numbers ξ , η , $p - \xi$, $p - \eta$.

We further observe that for no such number can $\xi = \eta$ hold; for then it would follow that

$$p(x + y) = a(\xi + \eta) + b(\xi - \eta) = 2a\xi,$$

which is absurd, since none of the factors 2, a , ξ is divisible by p . In a similar way the equation

$$p(x - y + a + b) = 2a\xi + (a + b)(p - \xi - \eta)$$

shows that $\xi + \eta = p$ cannot hold.

Therefore, since the numbers $\xi - \eta$ and $p - \xi - \eta$ must be either positive or negative, we derive from this a subdivision of the system S into four complexes C , C' , C'' , C''' , namely by putting into them as follows:

into the complex	the numbers for which
C	$\xi - \eta$ is positive, $p - \xi - \eta$ is positive
C'	$\xi - \eta$ is positive, $p - \xi - \eta$ is negative
C''	$\xi - \eta$ is negative, $p - \xi - \eta$ is negative
C'''	$\xi - \eta$ is negative, $p - \xi - \eta$ is positive

Thus the criterion for a number of the complex C is properly sixfold: the six numbers ξ , η , $p - \xi$, $p - \eta$, $\xi - \eta$, $p - \xi - \eta$ must be positive. But plainly conditions 2, 5, and 6 already imply the rest. Similar statements hold for the complexes C' , C'' , C''' , so that the complete criteria are triple:

for the complex	the numbers that must be positive
C	η , $\xi - \eta$, $p - \xi - \eta$
C'	$p - \xi$, $\xi - \eta$, $\xi + \eta - p$
C''	$p - \eta$, $\eta - \xi$, $\xi + \eta - p$
C'''	ξ , $\eta - \xi$, $p - \xi - \eta$

Moreover, even without our saying so, anyone will readily understand that, in the figurative representation of complex numbers (see Article 39), the numbers of the system S are contained inside the square whose sides join the points representing the numbers 0, $a + bi$, $(1 + i)(a + bi)$, and $i(a + bi)$, and that the subdivision of the system S corresponds to the partition of the square by its diagonal lines. But at this point we have preferred to use purely arithmetical reasonings, leaving the illustration through geometric intuition to the experienced reader for the sake of brevity.

Art. 70

If four complex numbers $r = x + yi$, $r' = x' + y'i$, $r'' = x'' + y''i$, $r''' = x''' + y'''i$ are connected with one another so that

$$r' = m + ir, \quad r'' = m + ir' = (1 + i)m - r, \quad r''' = m + ir'' = im - ir,$$

and the first, r , is supposed to belong to the complex C , then the remaining numbers r' , r'' , r''' will belong respectively to the complexes C' , C'' , C''' .

Indeed, putting $\xi = ax + by$, $\eta = ay - bx$, $\xi' = ax' + by'$, $\eta' = ay' - bx'$, $\xi'' = ax'' + by''$, $\eta'' = ay'' - bx''$, $\xi''' = ax''' + by'''$, $\eta''' = ay''' - bx'''$, one finds

$$\begin{aligned}\eta &= p - \xi' = p - \eta'' = \xi''', \\ \xi - \eta &= \xi' + \eta' - p = \eta'' - \xi'' = p - \xi''' - \eta''', \\ p - \xi - \eta &= \xi' - \eta' = \xi'' + \eta'' - p = \eta''' - \xi'''.\end{aligned}$$

The truth of the theorem then follows at once with the aid of the criteria.

And since again $r = m + ir'''$, it is easily seen that if r is supposed to belong to C' , then the numbers r' , r'' , r''' belong respectively to C'' , C''' , C ; if r belongs to C'' , they belong to C''' , C , C' ; and finally, if r belongs to C''' , they belong to C , C' , C'' . At the same time it follows that in each of the complexes C , C' , C'' , C''' there are equally many numbers, namely $\frac{1}{4}(p-1)$.

Art. 71

Theorem. Let k denote an integer not divisible by m . Multiply each number of the complex C by k , and distribute the simply least residues of the products modulo m among the complexes C , C' , C'' , C''' . If the numbers of those residues which belong to the four complexes are denoted respectively by c , c' , c'' , c''' , then the character of the number k with respect to the modulus m will be

$$\equiv c' + 2c'' + 3c''' \pmod{4}.$$

Proof. Let the c least residues belonging to C be α , β , γ , δ , etc.; then let the c' residues belonging to C' be $m + i\alpha'$, $m + i\beta'$, $m + i\gamma'$, $m + i\delta'$, etc.; further, let the c'' residues belonging to C'' be $(1+i)m - \alpha''$, $(1+i)m - \beta''$, $(1+i)m - \gamma''$, $(1+i)m - \delta''$, etc.; and finally let the c''' residues belonging to C''' be $im - i\alpha'''$, $im - i\beta'''$, $im - i\gamma'''$, $im - i\delta'''$, etc.

Now consider four products, namely:

1. the product of all the $\frac{1}{4}(p-1)$ numbers that constitute the complex C ;
2. the product of the products obtained by multiplying each of these numbers by k ;
3. the product of the least residues of these products, namely of the numbers α , β , γ , δ , etc., $m + i\alpha'$, $m + i\beta'$, etc. etc.;
4. the product of all the $c + c' + c'' + c'''$ numbers α , β , γ , δ , etc., α' , β' , γ' , etc., α'' , β'' , γ'' , etc., α''' , β''' , γ''' , etc.

Denote these four products, in order, by P , P' , P'' , P''' . Plainly

$$P' = k^{\frac{1}{4}(p-1)}P, \quad P' \equiv P'', \quad P'' \equiv P''' i^{c'+2c''+3c'''} \pmod{m},$$

and hence

$$Pk^{\frac{1}{4}(p-1)} \equiv P''' i^{c'+2c''+3c'''} \pmod{m}.$$

But it is easily seen that the numbers $\alpha', \beta', \gamma', \text{etc.}, \alpha'', \beta'', \gamma'', \text{etc.}, \alpha''', \beta''', \gamma''', \text{etc.}$ all belong to the complex C , and that they are distinct both from one another and from the numbers $\alpha, \beta, \gamma, \delta, \text{etc.}$, just as these latter are mutually distinct. All these numbers taken together, and abstracting from their order, must therefore be completely identical with all the numbers constituting the complex C . Hence we infer that $P = P'''$, and so

$$Pk^{\frac{1}{4}(p-1)} \equiv Pi^{c'+2c''+3c'''} \pmod{m}.$$

Finally, since none of the individual factors of the product P is divisible by m , it follows that

$$k^{\frac{1}{4}(p-1)} \equiv i^{c'+2c''+3c'''} \pmod{m},$$

and hence $c' + 2c'' + 3c'''$ is the character of the number k with respect to the modulus m . Q. E. D.

Art. 72

In order that the general theorem of the preceding article may be applied to the number $1 + i$, the complex C must again be subdivided into two smaller complexes G and G' . We shall refer to G those numbers $x + yi$ for which $ax + by = \xi$ is less than $\frac{1}{2}p$, and to G' those for which ξ is greater than $\frac{1}{2}p$. We shall denote by g and g' respectively the number of entries contained in G and G' , whence $g + g' = \frac{1}{4}(p - 1)$.

The complete criterion for numbers belonging to G will therefore be that the three numbers $\eta, \xi - \eta, p - 2\xi$ are positive; for the third condition for the complex C , according to which $p - \xi - \eta$ must be positive, is already implicitly contained in these, since

$$p - \xi - \eta = (\xi - \eta) + (p - 2\xi).$$

Similarly, the complete criterion for numbers belonging to G' consists in positive values of the three numbers $\eta, p - \xi - \eta$, and $2\xi - p$.

It follows easily that the product of any number of the complex G by the number $1 + i$ belongs to the complex C''' . For if we put

$$\begin{aligned} (x + yi)(1 + i) &= x' + y'i, & ax' + by' &= \xi', & ay' - bx' &= \eta', \\ \text{one finds} \quad \xi' &= \xi - \eta, & \eta' - \xi' &= 2\eta, & p - \xi' - \eta' &= p - 2\xi, \end{aligned}$$

that is, the criterion for a number $x + yi$ placed in the complex G is identical with the criterion for the number $x' + y'i$ belonging to the complex C''' . In exactly the same way it is shown that the product of any number of G' by $1 + i$ belongs to the complex C'' .

Therefore, if in the preceding article we give k the value $1 + i$, then $c = 0, c' = 0, c'' = g', c''' = g$, and hence the character of the number $1 + i$ becomes

$$3g + 2g' = \frac{1}{2}(p - 1) + g.$$

And since the characters of the numbers i and -1 are $\frac{1}{4}(p - 1)$ and $\frac{1}{2}(p - 1)$, respectively, the characters of the numbers $-1 + i, -1 - i, 1 - i$ are respectively

$$\frac{3}{4}(p - 1) + g, \quad g, \quad \frac{1}{4}(p - 1) + g.$$

Thus the whole crux of the matter now turns on the investigation of the number g .

Art. 73

What we have set out in Articles 69–72 is properly independent of the supposition that m is a primary number. From now on, however, we shall at least suppose that a is odd, b even, and moreover that a , b , and $a - b$ are positive numbers.

Above all, the limits of the values of x in the complex G must be established. Putting

$$ay - bx = \eta, \quad (a + b)x - (a - b)y = \zeta, \quad p - 2ax - 2by = \theta,$$

the criterion for numbers $x + yi$ belonging to the complex G consists in the three conditions that η , ζ , and θ are positive numbers.

Since

$$px = (a - b)\eta + a\zeta, \quad p(a - 2x) = a\theta + 2b\eta,$$

it is plain that x and $2a - x$ must be positive numbers; that is, x must be equal to one of the numbers $1, 2, 3, \dots, \frac{1}{2}(a - 1)$.

Further, since

$$(a - b)\theta = 2b\zeta + p(a - b - 2x),$$

it is clear that as long as x is less than $\frac{1}{2}(a - b)$, the second condition (according to which ζ must be positive) already implies the third (that θ must be positive); on the other hand, whenever x is greater than $\frac{1}{2}(a - b)$, the second condition is already contained under the third. Therefore, for the values of x equal to $1, 2, 3, \dots, \frac{1}{2}(a - b - 1)$, one need only ensure that η and ζ become positive; that is, that y is greater than $\frac{bx}{a}$ and less than $\frac{(a+b)x}{a-b}$. Thus, for such a given value of x , there will be exactly

$$\left[\frac{(a + b)x}{a - b} \right] - \left[\frac{bx}{a} \right]$$

numbers $x + yi$, if we use brackets in the same meaning in which we have already used them in many other places.

On the other hand, for the values of x equal to $\frac{1}{2}(a - b + 1), \frac{1}{2}(a - b + 3), \dots, \frac{1}{2}(a - 1)$, it will suffice to make η and θ positive; that is, y must be greater than $\frac{bx}{a}$ and less than

$$\frac{p - 2ax}{2b} = \frac{1}{2}b + \frac{aa - 2ax}{2b}.$$

Hence, for such a given value of x , there will be exactly

$$\left[\frac{1}{2}b + \frac{aa - 2ax}{2b} \right] - \left[\frac{bx}{a} \right]$$

numbers $x + yi$. We therefore infer that the number of entries of the complex G is

$$g = \Sigma \left[\frac{(a + b)x}{a - b} \right] + \Sigma \left[\frac{1}{2}b + \frac{aa - 2ax}{2b} \right] - \Sigma \left[\frac{bx}{a} \right],$$

where in the first term the summation is to be extended over all integral values of x from 1 to $\frac{1}{2}(a - b - 1)$; in the second from $\frac{1}{2}(a - b + 1)$ to $\frac{1}{2}(a - 1)$; in the third from 1 to $\frac{1}{2}(a - 1)$.

If we use the characteristic φ in the same meaning as in the cited place, namely

$$\varphi(t, u) = \left[\frac{u}{t} \right] + \left[\frac{2u}{t} \right] + \left[\frac{3u}{t} \right] + \cdots + \left[\frac{t'u}{t} \right],$$

where t and u denote any positive numbers and t' denotes the number $\left[\frac{1}{2}t \right]$, then the first term becomes $\varphi(a - b, a + b)$, and the third becomes $-\varphi(a, b)$. The second becomes

$$\frac{1}{4}bb + \sum \left[\frac{aa - 2ax}{2b} \right].$$

But, writing the terms in the reverse order,

$$\Sigma \left[\frac{aa - 2ax}{2b} \right] = \left[\frac{a}{2b} \right] + \left[\frac{3a}{2b} \right] + \left[\frac{5a}{2b} \right] + \cdots + \left[\frac{(b-1)a}{2b} \right] = \varphi(2b, a) - \varphi(b, a).$$

Thus our formula takes the form

$$g = \varphi(a - b, a + b) + \varphi(2b, a) - \varphi(a, b) - \varphi(b, a) + \frac{1}{4}bb.$$

First consider the term $\varphi(a - b, a + b)$, which is at once transformed into

$$\varphi(a - b, 2b) + 1 + 2 + 3 + \cdots + \frac{1}{2}(a - b - 1),$$

or into

$$\varphi(a - b, 2b) + \frac{1}{8}((a - b)^2 - 1).$$

Next, since by the general theorem

$$\varphi(t, u) + \varphi(u, t) = \left[\frac{1}{2}t \right] \cdot \left[\frac{1}{2}u \right],$$

when t and u are positive relatively prime integers, we have

$$\varphi(a - b, 2b) = \frac{1}{2}b(a - b - 1) - \varphi(2b, a - b),$$

and therefore

$$\varphi(a - b, a + b) = \frac{1}{8}(aa + 2ab - 3bb - 4b - 1) - \varphi(2b, a - b).$$

Let us arrange the parts of $\varphi(2b, a - b)$ as follows:

$$\begin{aligned} & \left[\frac{a - b}{2b} \right] + \left[\frac{3(a - b)}{2b} \right] + \left[\frac{5(a - b)}{2b} \right] + \cdots + \left[\frac{(b-1)(a - b)}{2b} \right] \\ & + \left[\frac{a - b}{b} \right] + \left[\frac{2(a - b)}{b} \right] + \left[\frac{3(a - b)}{b} \right] + \cdots + \left[\frac{\frac{1}{2}b(a - b)}{b} \right]. \end{aligned}$$

The second series plainly becomes

$$\varphi(b, a - b) = \varphi(b, a) - 1 - 2 - 3 - \dots - \frac{1}{2}b = \varphi(b, a) - \frac{1}{8}(bb + 2b).$$

We write the first series in the reverse order of its terms as

$$\begin{aligned} & \left[\frac{1}{2}(a + 1 - b) - \frac{a}{2b} \right] + \left[\frac{1}{2}(a + 3 - b) - \frac{3a}{2b} \right] \\ & + \left[\frac{1}{2}(a + 5 - b) - \frac{5a}{2b} \right] + \dots + \left[\frac{1}{2}(a - 1) - \frac{(b - 1)a}{2b} \right], \end{aligned}$$

which expression, since in general, if t denotes an integer and u a fraction, $[t - u] = t - 1 - [u]$, is changed into

$$\begin{aligned} & \frac{1}{8}b(2a - 4 - b) - \left[\frac{a}{2b} \right] - \left[\frac{3a}{2b} \right] - \left[\frac{5a}{2b} \right] - \dots - \left[\frac{(b - 1)a}{2b} \right] \\ & = \frac{1}{8}b(2a - 4 - b) - \varphi(2b, a) + \varphi(b, a). \end{aligned}$$

Hence

$$\varphi(2b, a - b) = 2\varphi(b, a) - \varphi(2b, a) + \frac{1}{4}b(a - 3 - b),$$

and consequently

$$\varphi(a - b, a + b) = \varphi(2b, a) - 2\varphi(b, a) + \frac{1}{8}(aa - bb + 2b - 1).$$

Substituting this value in the formula for g given above, and also using

$$\varphi(a, b) + \varphi(b, a) = \frac{1}{4}b(a - 1),$$

we obtain

$$g = 2\varphi(2b, a) - 2\varphi(b, a) + \frac{1}{8}(aa - 2ab + bb + 4b - 1).$$

Art. 74

By entirely similar reasonings one handles the case in which a and b remain positive but $a - b$ is negative, or $b - a$ positive. The equations

$$p(a - 2x) = 2b\eta + a\theta, \quad p(b - a + 2x) = 2b\zeta + (b - a)\theta$$

show that $\frac{1}{2}a - x$ and $x + \frac{1}{2}(b - a)$ are positive, and hence that x must be equal to one of the numbers

$$-\frac{1}{2}(b - a - 1), -\frac{1}{2}(b - a - 3), -\frac{1}{2}(b - a - 5), \dots, +\frac{1}{2}(a - 1).$$

Further, from the equation $px + (b - a)\eta = a\zeta$ it follows that, for negative values of x , the condition by which η must be positive is already contained under the condition by which ζ

must be positive; the opposite happens whenever a positive value is assigned to x . Hence the values of y , for a fixed negative value of x , must be contained between $\frac{(a+b)x}{a-b}$ and $\frac{p-2ax}{2b}$, while for a positive value of x they must be contained between $\frac{bx}{a}$ and $\frac{p-2ax}{2b}$. Plainly for $x = 0$ these limits are 0 and $\frac{p-2ax}{2b}$, the value $y = 0$ itself being excluded.

It follows that

$$g = - \sum \left[\frac{(a+b)x}{a-b} \right] + \sum \left[\frac{1}{2}b + \frac{aa-2ax}{2b} \right] - \sum \left[\frac{bx}{a} \right],$$

where in the first term the summation is to be extended over all negative values of x from -1 to $-\frac{1}{2}(b-a-1)$; in the second over all values of x from $-\frac{1}{2}(b-a-1)$ to $\frac{1}{2}(a-1)$; in the third over all positive values of x from $+1$ to $\frac{1}{2}(a-1)$. In this way the first summation gives $-\varphi(b-a, b+a)$; the second, as in the preceding article, gives $\frac{1}{4}bb + \varphi(2b, a) - \varphi(b, a)$; and the third gives $-\varphi(a, b)$. Thus

$$g = -\varphi(b-a, b+a) + \varphi(2b, a) - \varphi(b, a) - \varphi(a, b) + \frac{1}{4}bb.$$

Now, in the same way as in the preceding article,

$$\begin{aligned} \varphi(b-a, b+a) &= \varphi(b-a, 2b) - \frac{1}{8}((b-a)^2 - 1) \\ &= \frac{1}{8}(3bb - 2ab - aa - 4b + 1) - \varphi(2b, b-a), \end{aligned}$$

and also

$$\varphi(2b, b-a) = \varphi(2b, a) - 2\varphi(b, a) + \frac{1}{4}b(b-1-a),$$

so that

$$\varphi(b-a, b+a) = 2\varphi(b, a) - \varphi(2b, a) + \frac{1}{8}(bb - aa - 2b + 1),$$

and finally

$$g = 2\varphi(2b, a) - 2\varphi(b, a) + \frac{1}{8}(aa - 2ab + bb + 4b - 1).$$

It has therefore been established that the same formula for g holds whether $a-b$ is positive or negative, provided a and b are positive.

Art. 75

To obtain the further reduction, put

$$\begin{aligned} L &= \left[\frac{a}{2b} \right] + \left[\frac{2a}{2b} \right] + \left[\frac{3a}{2b} \right] + \cdots + \left[\frac{\frac{1}{2}ba}{2b} \right], \\ M &= \left[\frac{(\frac{1}{2}b+1)a}{2b} \right] + \left[\frac{(\frac{1}{2}b+2)a}{2b} \right] + \left[\frac{(\frac{1}{2}b+3)a}{2b} \right] + \cdots + \left[\frac{ba}{2b} \right], \\ N &= \left[\frac{a+b}{2b} \right] + \left[\frac{2a+b}{2b} \right] + \left[\frac{3a+b}{2b} \right] + \cdots + \left[\frac{\frac{1}{2}ba+b}{2b} \right]. \end{aligned}$$

Since it is easily seen that, in general, $[u] + [u + \frac{1}{2}] = [2u]$ for any real quantity u , we get $L + N = \varphi(b, a)$; and since plainly $L + M = \varphi(2b, a)$, we have

$$\varphi(2b, a) - \varphi(b, a) = M - N.$$

Further, it is obvious that the sum of the first term of the series N with the penultimate term of the series M , namely

$$\left[\frac{a+b}{2b} \right] + \left[\frac{(b-1)a}{2b} \right],$$

becomes $\frac{1}{2}(a-1)$, and that the same sum is produced from the second term of the series N with the antepenultimate term of M , and so on. Hence, since also the last term of the series M becomes $\frac{1}{2}(a-1)$, while the last term of the series N is

$$\left[\frac{a+2}{4} \right] = \frac{1}{4}(a \mp 1),$$

with the upper or lower sign holding according as a is of the form $4n+1$ or $4n-1$, we get

$$M + N = \frac{1}{4}(a-1)b + \frac{1}{4}(a \mp 1),$$

and hence

$$\varphi(2b, a) - \varphi(b, a) = \frac{1}{4}(a-1)b + \frac{1}{4}(a \mp 1) - 2N.$$

Thus the formula for g found in Articles 73 and 74 passes into

$$g = \frac{1}{8}((a+b)^2 - 1) + 2n - 4N,$$

putting $a \mp 1 = 4n$, where n is an integer. But since from this we have $1 = 16nn - 8an + aa$, the formula may also be written in the following way:

$$g = \frac{1}{8}(-aa + 2ab + bb + 1) + 4 \left(\frac{1}{2}(a+1)n - nn - N \right).$$

Consequently, since g is the character of the number $-1-i$ with respect to the modulus $a+bi$, this character becomes

$$\equiv \frac{1}{8}(-aa + 2ab + bb + 1) \pmod{4}.$$

This is precisely the theorem found above by induction in Article 64, and from it the theorems about the characters of the numbers $1+i$, $1-i$, and $-1+i$ follow at once. Therefore these four theorems have now been rigorously proved for the case in which a and b are positive.

Art. 76

If, while a remains positive, b is negative, put $b = -b'$, so that b' is positive. Since it has now been proved that, for the modulus $a+b'i$, the character of the number $-1-i$ is

$$\equiv \frac{1}{8}(-aa + 2ab' + b'b' + 1) \pmod{4},$$

the character of the number $-1 + i$ for the modulus $a - b'i$, by the theorem stated in Article 62, will be

$$\equiv \frac{1}{8}(aa - 2ab' - b'b' - 1).$$

That is, the character of the number $-1 + i$ for the modulus $a + bi$ becomes

$$\equiv \frac{1}{8}(aa + 2ab - bb - 1).$$

But this is exactly the theorem stated in Article 64, from which the remaining three statements about the characters of $1 + i$, $1 - i$, and $-1 - i$ follow at once. Therefore these theorems have also been proved for the case in which b is negative, that is, for all cases in which a is positive.

Finally, if a is negative, put $a = -a'$, $b = -b'$. Since by what has already been proved the character of the number $1 + i$ with respect to the modulus $a' + b'i$ is

$$\equiv \frac{1}{8}(-a'a' + 2a'b' - 3b'b' + 1) \pmod{4},$$

and since it makes no difference whether we take $a' + b'i$ or its opposite $-a' - b'i$ as modulus, it is plain that the character of the number $1 + i$ with respect to the modulus $a + bi$ is

$$\equiv \frac{1}{8}(-aa + 2ab - 3bb + 1),$$

and similar statements hold about the characters of the numbers $1 - i$, $-1 + i$, and $-1 - i$. It follows from this that the proof of the theorems about the characters of the numbers $1 + i$, $1 - i$, $-1 + i$, and $-1 - i$ (Articles 63 and 64) is no longer subject to any limitation.